6. Create a sign in page which takes username and password, a checkbox to remember username and a button. Store the username using shared preference when check box is checked. When you press the sign in button it has to display the success page if the credentials are correct. Also provide the facility to enter wrong attempts at most 3 times. Show the remaining attempts using alert dialog box. After 3 wrong attempts it has to disable sign in button. If the credentials are stored already then, the page has to show the credentials.

MainActivity.java:

```
package com.example.loginapp;

import android.content.Context;
import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.EditText;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    EditText user, pass;
    CheckBox rem;
    Button btn;
    int att = 3;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
        user = findViewById(R.id.edtusername);
        pass = findViewById(R.id.edtpassword);
        rem = findViewById(R.id.chkremember);
        btn = findViewById(R.id.btnsignin);
```

```
    SharedPreferences sp =getSharedPreferences("login", Context.MODE_PRIVATE);
    String name = sp.getString("username", "");
    if (!name.equals("")) {
      user.setText(name);
    }
    btn.setOnClickListener(new View.OnClickListener() {

      @Override
      public void onClick(View v) {
        String u = user.getText().toString();
        String p = pass.getText().toString();

        String su = sp.getString("username", "");
        String spass = sp.getString("password", "");

        if (rem.isChecked()) {
          SharedPreferences.Editor ed = sp.edit();
          ed.putString("username", u);
          ed.putString("password", p);
          ed.commit();
          openSuccess(u);
        }
        else {
          if (!su.equals("")) {
            if (u.equals(su) && p.equals(spass)) {
              openSuccess(u);
            } else {
              att--;
              showAlertBox();
              if (att == 0) {
                btn.setEnabled(false);
              }
            }
          } else {
            openSuccess(u);
          }
        }
      }
    });
  }

  private void openSuccess(String u) {
    Intent i =new Intent(MainActivity.this, Success.class);
    i.putExtra("name", u);
    startActivity(i);
```

```
    }

    private void showAlertBox() {
        AlertDialog.Builder bld =new AlertDialog.Builder(MainActivity.this);
        bld.setTitle("Wrong Credentials")
                .setMessage("Remaining attempts: " + att)
                .setPositiveButton("OK", null);
        AlertDialog dlg = bld.create();
        dlg.show();
    }
}
```

Success.java:

```
package com.example.loginapp;

import android.content.Intent;
import android.os.Bundle;
import android.widget.TextView;

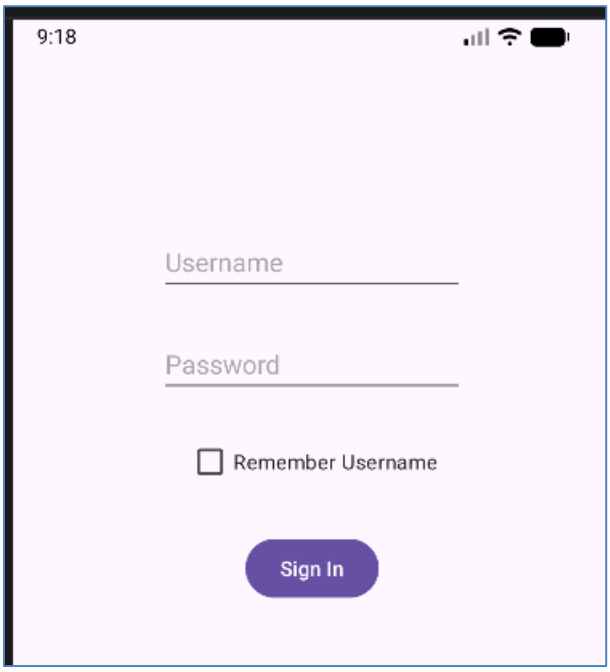
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Success extends AppCompatActivity {
    TextView tv;

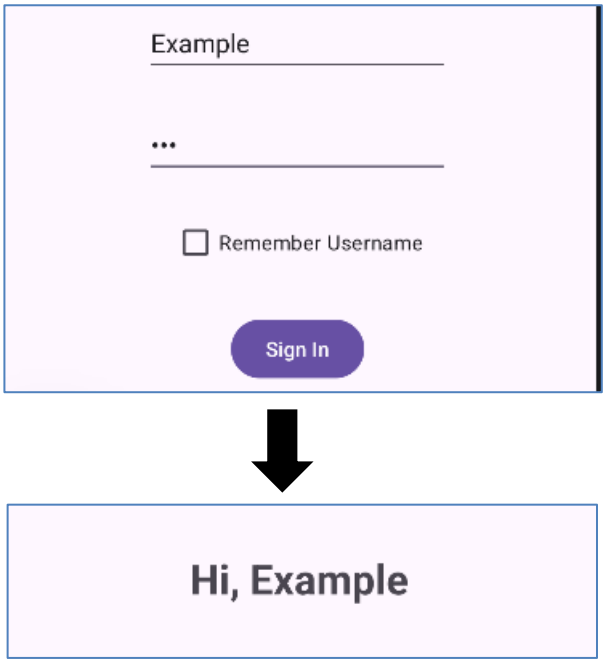
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_success);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
        tv = findViewById(R.id.tv);
        Intent i = getIntent();
        String name = i.getStringExtra("name");
        tv.setText("Hi, " + name);
    }
}
```

OUTPUT:

1. Sign-in page — initial screen



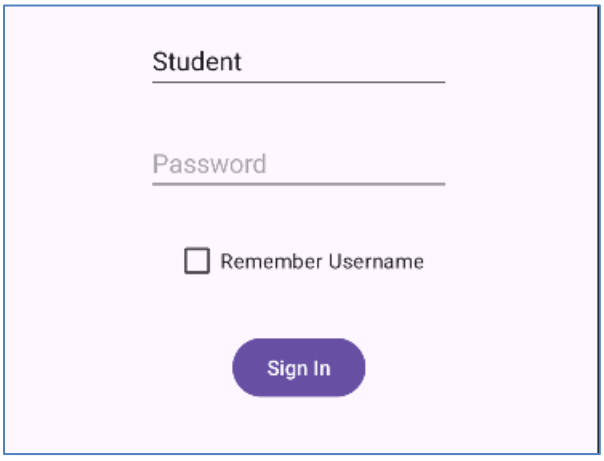
2. First login without checkbox

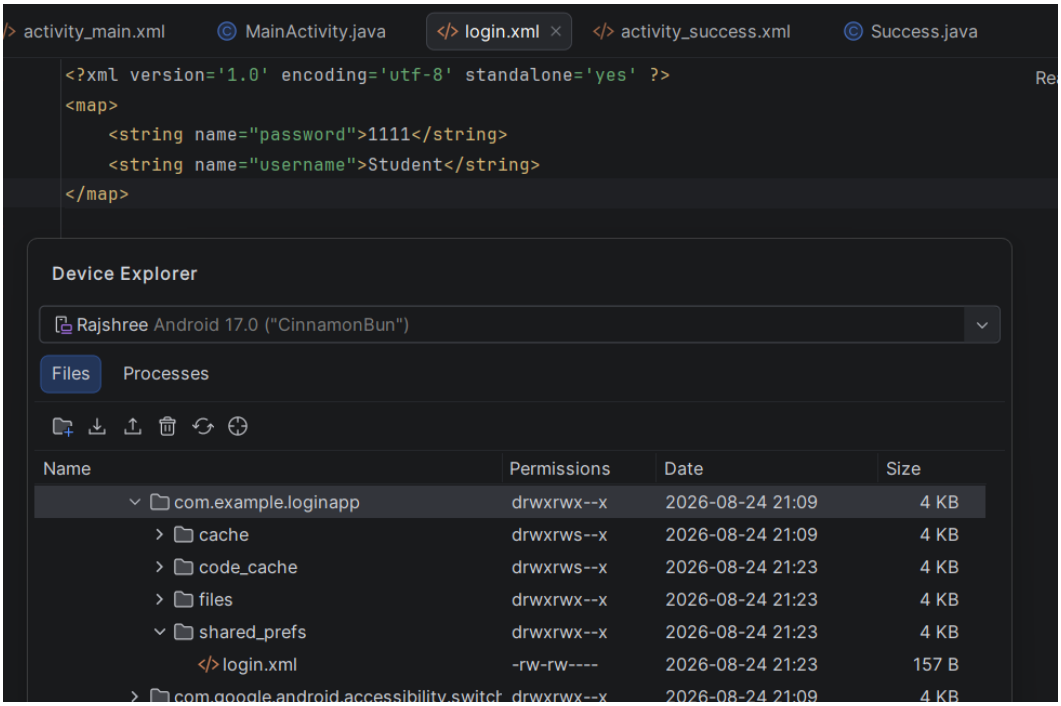


3. Login with Remember Username checked

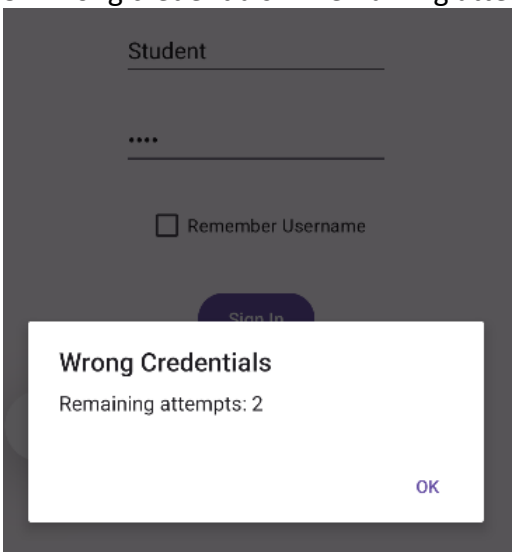


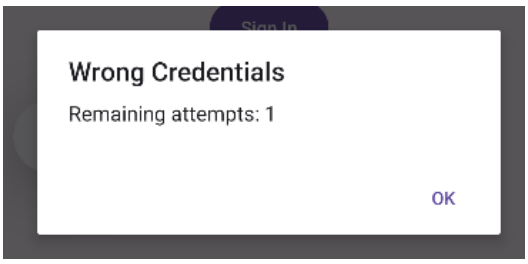
4. Stored username appearing

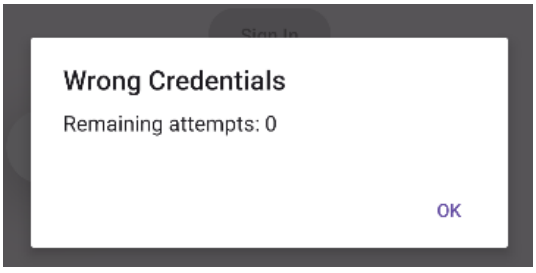




5. Wrong credentials — remaining attempts







6. Three wrong attempts / button disabled

